

Q:- Find a basis of the Column space, row space and null space of the matrix

$$A = \begin{bmatrix} 1 & 1 & 5 & 1 \\ 2 & 4 & 14 & 4 \\ 2 & 3 & 12 & 3 \end{bmatrix}$$

OR find $\text{col}(A)$, $\text{row}(A)$, $\text{Null}(A)$.

OR find $C(A)$, $R(A)$, $N(A)$.

Solution, ① Given matrix is

$$A = \begin{bmatrix} 1 & 1 & 5 & 1 \\ 2 & 4 & 14 & 4 \\ 2 & 3 & 12 & 3 \end{bmatrix}$$

By $R_2 - 2R_1$, $R_3 - 2R_1$

$$\sim \begin{bmatrix} 1 & 1 & 5 & 1 \\ 0 & 2 & 4 & 2 \\ 0 & 1 & 2 & 1 \end{bmatrix}$$

By R_{23}

$$\sim \begin{bmatrix} 1 & 1 & 5 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 2 & 4 & 2 \end{bmatrix}$$

NOTE..

For row space
reduce given
matrix into
reduced echelon
form

By $R_1 - R_2, R_3 - 2R_2$

$$R \sim \begin{bmatrix} 1 & 0 & 3 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Basis for row space $R(A)$ is

$$\{(1, 0, 3, 0), (0, 1, 2, 1)\}$$

and $\dim R(A) = 2$ or $\text{Rank}(A) = 2$.

(ii). Given

$$A = \begin{bmatrix} 1 & 1 & 5 & 1 \\ 2 & 4 & 14 & 4 \\ 2 & 3 & 12 & 3 \end{bmatrix}$$

Here

$$A^t = \begin{bmatrix} 1 & 2 & 2 \\ 1 & 4 & 3 \\ 5 & 14 & 12 \\ 1 & 4 & 3 \end{bmatrix}$$

NOTE

For column space
reduce transpose
of given matrix
into reduced
echelon form

By $R_2 - R_1, R_3 - 5R_1, R_4 - R_1$

$$R \sim \begin{bmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ 0 & 4 & 2 \\ 0 & 2 & 1 \end{bmatrix}$$

By $R_1 - R_2, R_3 - 2R_2, R_4 - R_2$

$$R \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Basis for column space $\text{Col}(A)$ is

$$\{(1, 0, 1), (0, 2, 1)\} \text{ and } \dim(\text{Col}(A)) = 2.$$

(iii). Given

$$A = \begin{bmatrix} 1 & 1 & 5 & 1 \\ 2 & 4 & 14 & 4 \\ 2 & 3 & 12 & 3 \end{bmatrix}$$

Consider

$$Ax = 0$$

$$\begin{bmatrix} 1 & 1 & 5 & 1 \\ 2 & 4 & 14 & 4 \\ 2 & 3 & 12 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} x_1 + x_2 + 5x_3 + x_4 \\ 2x_1 + 4x_2 + 14x_3 + 4x_4 \\ 2x_1 + 3x_2 + 12x_3 + 3x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

we get

$$x_1 + x_2 + 5x_3 + x_4 = 0$$

$$2x_1 + 4x_2 + 14x_3 + 4x_4 = 0$$

$$2x_1 + 3x_2 + 12x_3 + 3x_4 = 0$$

The augmented matrix is

$$A_b = \begin{bmatrix} 1 & 1 & 5 & 1 & : & 0 \\ 2 & 4 & 14 & 4 & : & 0 \\ 2 & 3 & 12 & 3 & : & 0 \end{bmatrix}$$

By $R_2 - 2R_1$, $R_3 - 2R_1$

$$\sim \begin{bmatrix} 1 & 1 & 5 & 1 & : & 0 \\ 0 & 2 & 4 & 2 & : & 0 \\ 0 & 1 & 2 & 1 & : & 0 \end{bmatrix}$$

By R_{23}

$$\sim \begin{bmatrix} 1 & 1 & 5 & 1 & : & 0 \\ 0 & 1 & 2 & 1 & : & 0 \\ 0 & 2 & 4 & 2 & : & 0 \end{bmatrix}$$

NOTE:

Null space of an $m \times n$ matrix A is the set of all solutions to the homogeneous equation $Ax = 0$

where x is a column matrix of order n .

By $R_1 - R_2, R_3 - 2R_2$

$$\sim R \begin{bmatrix} 1 & 0 & 3 & 0 & | & 0 \\ 0 & 1 & 2 & 1 & | & 0 \\ 0 & 0 & 0 & 0 & | & 0 \end{bmatrix}$$

The equivalent system is

$$x_1 + 3x_3 = 0$$

$$\text{and } x_2 + 2x_3 + x_4 = 0$$

$$\Rightarrow x_1 = -3x_3$$

$$\text{and } x_2 = -2x_3 - x_4$$

Now

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -3x_3 \\ -2x_3 - x_4 \\ x_3 \\ x_4 \end{bmatrix}$$

$$= \begin{bmatrix} -3x_3 \\ -2x_3 \\ x_3 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -x_4 \\ 0 \\ x_4 \end{bmatrix}$$

$$= x_3 \begin{bmatrix} -3 \\ -2 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 0 \\ -1 \\ 0 \\ 1 \end{bmatrix}$$

Basis for null space is

$$\{(-3, -2, 1, 0), (0, -1, 0, 1)\}$$

$$\text{and } \dim(\text{Nul}(A)) = 2$$